Multi-Sport Collegiate Athletic Performance & Recovery Intelligence System

Agentic AI Project Proposal

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1. Problem Statement & Motivation

Collegiate athletic departments face critical challenges managing athlete health across 15-20 sports programs. Each sport operates in silos with athletic trainers, strength coaches, and medical staff working independently without systematic knowledge sharing. This leads to preventable injuries from inadequate load monitoring, inefficient use of sports medicine resources, duplicated protocol development efforts, and missed cross-sport learning opportunities. Target users include athletic trainers managing injury prevention across teams, strength coaches coordinating training programs, team physicians overseeing clearances, student-athletes needing personalized recovery guidance, and athletic directors managing resources. An agentic AI solution is uniquely appropriate because it requires continuous monitoring of dynamic, multi-dimensional data from diverse sources with real-time adaptation to changing conditions. The complexity of balancing individual athlete needs across all programs exceeds human cognitive capacity for optimization. Agents operate 24/7 to flag trends, suggest adjustments, and coordinate resources without manual communication delays. These are capabilities that fixed workflows or external parties cannot provide when a 6 PM injury requires immediate resource reallocation.

2. Use Cases & Scenarios

Scenario 1: Post-Surgical Recovery Optimization. A soccer goalkeeper recovering from ACL reconstruction needs safe return-to-play while maintaining fitness. Tasks: Recovery Protocol Agent monitors daily progress via wearables (ROM, gait, pain); Performance Analytics Agent identifies similar cases across sports and recommends evidence-based adjustments; Schedule Coordinator Agent books PT sessions and coordinates with strength coaches; Cross-Campus Aggregator benchmarks recovery timelines from other Division I programs; Injury Monitoring Agent flags deviations and alerts trainers immediately.

Scenario 2: Resource Optimization for Concurrent Injuries. An athletic trainer manages 15 injured athletes across five sports with overlapping schedules and limited PT resources. Tasks: Schedule Coordinator analyzes priority based on injury severity and competition schedules to create optimized daily schedules; resolves conflicts and ensures equitable equipment access (cryotherapy, underwater treadmills); coordinates physician clearances; Performance Analytics tracks treatment effectiveness and suggests modifications; system notifies trainer of potential overload situations.

Scenario 3: Cross-Campus Injury Prevention. A strength coach seeks to reduce hamstring strains after three recent cases in track & field. Tasks: Cross-Campus Aggregator searches 50+ institutions for low-injury programs; Performance Analytics analyzes their training loads, warm-ups, and strength benchmarks; Recovery Protocol reviews research via web search; Injury Monitoring continuously tracks current athletes' workload to flag elevated-risk individuals for preemptive intervention.

3. Initial System Design

Agent Architecture: Multi-agent system with specialized coordination. Injury Monitoring Agent ingests wearable data, medical records, injury reports to detect patterns. Recovery Protocol Agent maintains rehabilitation knowledge base with dynamic adjustments. Performance Analytics Agent processes training loads and biomechanics to identify risks. Schedule Coordinator Agent manages PT appointments, facilities, staff across programs. Cross-Campus Data Aggregator shares anonymized insights (HIPAA-compliant). Research Synthesis Agent performs continuous literature reviews via web search. **Tools & Data Sources:** Catapult Sports and TeamBuildr (training loads, GPS, force plates); HIPAA-compliant EHR systems (injury histories, clearances); Google Calendar and TeamSnap (scheduling); PubMed web search (research); campus facility management systems (equipment availability); Whoop and Polar wearables (sleep, HRV, recovery); secure cross-campus platform (anonymized benchmarking, FERPA-compliant).

4. Feasibility & Risks

Challenges: Medical hallucination risks requiring validation layers and human oversight; integration complexity across disparate APIs; long context needs for athlete histories causing performance issues; API downtime requiring fallback mechanisms; staff resistance to workflow changes. **Constraints:** HIPAA/FERPA compliance for data handling; API rate limits restricting real-time monitoring; computational costs for long-context interactions; legal barriers for cross-institutional sharing requiring data use agreements; limited ground truth data reducing edge case accuracy; incomplete cross-campus participation biasing insights. **Timeline (14 weeks):** Weeks 1-2: requirements gathering and architecture design via stakeholder interviews. Weeks 3-5: core agent development and API integration. Weeks 6-8: testing with historical data and simulated scenarios. Weeks 9-11: limited pilot with 1-2 sports programs. Weeks 12-13: refinement based on feedback and optimization. Week 14: final documentation and live demonstration.